Ahmad Samim Hamza

Senior Software & AI Engineer

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ABOUT ME

AI & Senior Software Engineer and Master's student in Artificial Intelligence at the University of Jyväskylä, with 5 years of experience building intelligent applications and the reliable software systems around them. At Konecranes, I owned the end-to-end delivery of an enterprise RAG and multi-agent assistant using AWS Bedrock (Agents, AgentCore), LangGraph, and LangChain—evaluating architectural trade-offs, integrating PostgreSQL feedback pipelines, and deploying on AWS (Amplify, ECS, RDS) with CloudWatch monitoring. Earlier in my career, I architected and delivered production ERP, logistics, and e-commerce platforms and led a team of eight engineers. I combine strong software engineering foundations (Python, FastAPI, TypeScript, NestJS, Next.js, PostgreSQL, Docker, CI/CD) with daily AI-assisted engineering workflows using Claude Code and Cursor to ship tested, production-grade code rapidly.

SKILLS

AI, ML & Agentic Systems: LangGraph | LangChain | AWS Bedrock (Agents & AgentCore) | Multi-Agent Orchestration | MCP | RAG Architectures & Vector Search (S3 Vector Store, ChromaDB) | Model Evaluation, Guardrails & LLMOps | Multi-Objective Optimization | PyTorch & scikit-learn | Computer Vision & ML Fundamentals

AI-Assisted Engineering: Claude Code | Cursor | Antigravity | Prompt & Spec Engineering

Backend & Data Engineering: Python (FastAPI, AsyncIO, Django) | TypeScript/JavaScript (NestJS, Node.js, Express) | PostgreSQL (Schema Design, Query Optimization, pgvector) | Semantic Web (Ontologies, RDF, OWL, SPARQL) | REST APIs | PHP (Laravel) | MySQL | SQLite | Redis

Frontend: Next.js | React | TypeScript | Tailwind CSS | Material UI | Vue.js | Nuxt.js

Cloud, Platforms & DevOps: AWS (Bedrock, ECS, Amplify, RDS, S3, Lambda, API Gateway, CloudWatch, ECR) | Azure (AI Foundry, AI Search) | Docker | Kubernetes | CI/CD (GitHub Actions) | Linux

WORK EXPERIENCE

Software Development Specialist (AI/ML) – Fixed-Term Summer Contract 05/2026 – 08/2026
Konecranes Hyvinkää, Finland

- Owned end-to-end design, development, and AWS deployment of a production enterprise AI assistant, taking the solution from initial problem discovery and prototyping to full documentation and handover to the internal team.
- Built and benchmarked multiple production-ready RAG variants on AWS Bedrock (Agents & AgentCore), LangChain, and LangGraph over 100+ internal XML, PDF, and XLSX documents indexed in an S3 vector store, evaluating trade-offs in retrieval accuracy, response latency, and maintainability.
- Architected a multi-agent system on AWS Bedrock with a supervisor agent orchestrating task-specific collaborator agents across enterprise knowledge sources.
- Integrated the RAG and agentic backends into a unified Next.js and FastAPI application behind a shared conversational UI, using PostgreSQL (AWS RDS) for user accounts, session management, and a human-in-the-loop feedback pipeline to evaluate three RAG variants across internal teams.
- Developed the entire Next.js and FastAPI application using Claude Code and Cursor as primary AI-assisted coding environments, accelerating iteration speed while enforcing strict production code quality.
- Deployed the platform on AWS (Amplify, ECS, RDS), collaborated with stakeholders to align agentic workflows with business operations, and monitored post-deployment health via CloudWatch logs and application error tracking to identify and resolve issues affecting retrieval accuracy and response latency.

Senior Software Engineer 05/2024 – 07/2025
Ramaki Group of Companies Kabul, Afghanistan

- Led and coordinated a team of eight engineers in the end-to-end development of Dara.af , a production e-commerce platform supporting hundreds of product listings and daily customer orders using Laravel, PostgreSQL, React, and Next.js.
- Designed and optimized PostgreSQL database schemas for complex transactional and inventory data.

- Built RESTful APIs and wrote unit, integration, and feature tests to ensure system reliability and data integrity.
- Conducted code reviews, established engineering standards, and mentored junior developers.

Software Engineer 09/2022 – 04/2024
Peace Global Logistics Kabul, Afghanistan
- Collaborated with logistics, finance, and operations domain experts to translate complex supply-chain and shipment tracking workflows into production ERP modules using NestJS, React, Next.js, and PostgreSQL.
- Led the architectural migration and database schema redesign of core ERP features (finance, logistics, inventory, and audit logging) into a modern NestJS, Next.js, and PostgreSQL stack serving 200+ daily users.
- Optimized complex PostgreSQL queries for high-volume order and shipment data and built reusable React + Tailwind dashboard components.

Instructor (Python & Machine Learning) 04/2022 – 10/2022
Kabul University Kabul, Afghanistan
- Taught Python programming and machine learning fundamentals to freshman and sophomore undergraduate students.

Full Stack Developer 07/2021 – 09/2022
Smart Friqi Kabul, Afghanistan
- Built production ERP modules using Laravel, Nuxt.js, and PostgreSQL; implemented real-time operational features using WebSockets; and developed interactive dashboards for business management and operations monitoring.

PROJECTS

Enterprise RAG & Multi-Agent Assistant (Konecranes) 05/2026 – 08/2026
Tech: AWS Bedrock (Agents, AgentCore), LangChain, LangGraph, FastAPI, Next.js, PostgreSQL, AWS Amplify/ECS/RDS, CloudWatch, Claude Code
- Internal enterprise knowledge assistant over 100+ heterogeneous documents (XML, PDF, XLSX), benchmarking multiple RAG variants behind a shared conversational UI with multi-agent supervisor orchestration, a PostgreSQL evaluation pipeline, and CloudWatch monitoring.

Dara.af E-Commerce Platform (dara.af) 05/2024 – 07/2025
Tech: Laravel, PostgreSQL, React, Next.js, REST API
- Led an 8-engineer team through PostgreSQL schema design, REST API development, and full-stack delivery for a production e-commerce platform supporting daily customer orders.

PGL ERP System (pglsystem.com) 09/2022 – 04/2024
Tech: NestJS, React, Next.js, PostgreSQL, REST API
- Built logistics, shipments, HR, finance, and inventory modules for Peace Global Logistics and designed APIs optimizing data flow across the enterprise platform.

Health Information RAG Chatbot (University of Jyväskylä) 2026
Tech: Python, ChromaDB, RAG, Vector Search, LLM Evaluation, Web Scraping
- Production-grade domain-specific RAG chatbot built by scraping public health information, chunking and indexing content into ChromaDB, and evaluating retrieval accuracy and generation quality using LLM evaluation frameworks.

EDUCATION

Master of Science in Artificial Intelligence 08/2025 – Present
University of Jyväskylä Jyväskylä, Finland
- Master's Thesis (In Progress): Exploring LLM-assisted validation for multi-objective optimization in Finnish forest planning.

Bachelor of Computer Science (Software Engineering) 2017 – 2021
Kabul University Kabul, Afghanistan

LANGUAGE SKILLS

- English (Fluent)
- Persian (Native)
- Finnish (A1 Level – Actively Learning)